

k-means and *k*-medoids

1. Start by writing **your own** *k*-means and *k*-medoids algorithms. For your own convenience, write your algorithms as a function subroutine. If you are coding in MATLAB, the structure of the *k*-medoids routine could be

```
function [I_assign,I_bar] = my_k_medoids(k,D,tau),
```

where the input consists of

k = number of clusters,
D = distance matrix of size $p \times p$,
tau = tolerance, a small number used for the stopping criterion (see the lecture notes),

and the output variables are

I_assign = assignment vector of length p , indicating the cluster of each data vector,
I_bar = index vector of length k , indicating the medoids.

To initialize the algorithm, include the following step:

- (i) Choose the initial clusters by picking randomly k data vectors from your data to be the initial medoids, computing the corresponding tightness.
 - (ii) Repeat the initialization n_{init} times (e.g., $n_{\text{init}} = 20$). This can be included as an input parameter if you wish), and pick the initial clustering that corresponds to the lowest tightness.
2. Test your *k*-means and *k*-medoids algorithm with $k = 3$ on the data file **WineData.mat** containing a data matrix $\mathbf{X} \in \mathbb{R}^{13 \times 178}$ of chemical analysis data of wines derived from the same area of Italy but originating from three different cultivars. The original data file can be found on the UCI Machine Learning Repository,

<https://archive.ics.uci.edu/ml/datasets/wine>

The attributes, corresponding to the rows of the data matrix $\mathbf{X}$ are concentrations/levels of the following substances:

- 1 Alcohol
- 2 Malic acid
- 3 Ash
- 4 Alcalinity of ash
- 5 Magnesium
- 6 Total phenols
- 7 Flavonoids
- 8 Nonflavonoid phenols
- 9 Proanthocyanins
- 10 Color intensity
- 11 Hue
- 12 OD280/OD315 of diluted wines
- 13 Proline

Comments on how well, or badly, your algorithms were able to cluster the data to correspond the three different cultivars by comparing your results with the true annotation recorded in the vector I included in the data file **WineData.mat**. Report whether the three wine types were easy to cluster based on the recorded attributes.

3. Download the data file **CardiacSPECT.mat**, originating from the UCI Machine Learning Repository site,

<https://archive.ics.uci.edu/ml/datasets/spect+heart>

The file contains a binary matrix X of size 22×187 and a vector I of length 187. The dataset reports 22 attributes extracted from cardiac Single Proton Emission Computed Tomography (SPECT) images of 187 patients. Each patient was classified into one of two categories: normal or abnormal. The database of 187 SPECT image sets (patients) was processed to extract features that summarize the original SPECT images. As a result, 44 continuous feature pattern was created for each patient. The pattern was further processed to obtain 22 binary feature patterns. The vector I , also in binary form, contains the annotation of the corresponding patient in one of the groups.

- (a) Write the distance matrix between the patients, using a dissimilarity index between the 0 and 1 as a distance measure.
- (b) Once you have the distance matrix, run the k -medoids algorithm to cluster the patients in two groups, A and B. To investigate how well - or badly - your clustering corresponds to the classification given by the cardiologist, write a matrix C of size 2×2 with the following entries:

$$\begin{aligned}c_{11} &= \text{number of 1's in your cluster A,} \\c_{12} &= \text{number of 1's in your cluster B,} \\c_{21} &= \text{number of 0's in your cluster A} \\c_{22} &= \text{number of 0's in your cluster B.}\end{aligned}$$

Comment on how the matrix C still provides you some insight on the degree of agreement between the clustering that you found and the classification by the cardiologist if

$$\begin{aligned}c_{11} &= \text{number of 1's in your cluster B,} \\c_{12} &= \text{number of 1's in your cluster A,} \\c_{21} &= \text{number of 0's in your cluster B} \\c_{22} &= \text{number of 0's in your cluster A.}\end{aligned}$$

In the light of your clustering, how well do the attributes represent the state of the patients?

4. This problem is similar to the previous one with a different data set. Download the attached Matlab data file **CongressionalVoteData.mat**, originating also from the UCI Machine Learning Repository,

<https://archive.ics.uci.edu/ml/datasets/congressional+voting+records>.

In your workspace, you find a data matrix X of size 16×435 , containing the votes of the 435 congressional representatives in 1984 on 16 issues, as well as a vector I of length 435 indicating their partisan membership of each representative (Republican = 0, Democrat = 1).

The columns of the Matrix X give a yes/no vote of each congressional representative on the following 16 issues (“yes” = 1, “no” = -1, “missing vote” = 0):

- 1 handicapped-infants
- 2 water-project-cost-sharing
- 3 adoption-of-the-budget-resolution
- 4 physician-fee-freeze
- 5 el-salvador-aid
- 6 religious-groups-in-schools
- 7 anti-satellite-test-ban
- 8 aid-to-Nicaraguan-contras
- 9 mx-missile
- 10 immigration
- 11 synfuels-corporation-cutback
- 12 education-spending
- 13 superfund-right-to-sue
- 14 crime
- 15 duty-free-exports
- 16 export-administration-act-South-Africa

- (a) Write the distance matrix between the representatives, using a dissimilarity index between the “yes” and “no” votes as a distance measure. To address the missing votes, you can follow the guidelines below:
- The data contain one representative who did not vote one single time. Discard that representative.
 - When computing the dissimilarity index between two representatives, include only the issues on which both of them voted. Hence, the dissimilarity index is the number of times the votes disagreed divided by all votes that both candidates cast.
 - You will find some pairs of representatives with no simultaneous voting record: one or the other was always absent. Use a neutral value $1/2$ as a dissimilarity measure for those pairs.
- (b) Once you have the distance matrix, run the k -medoids algorithm to cluster the votes in two groups. Then investigate if your clustering corresponds to the party line. You can use a similar matrix C as in the previous problem to summarize the clustering properties of the algorithm. In the light of your clustering, what can you say about the congress of 1984, in particular, how partisan the votes were? How did the two medoids representing their group vote? How was their absentee record? Discuss the suggested process of dealing with the missing votes. Could the missing vote be informative?
5. Download the iris data `IrisData.mat`, described in the PCA project sheet. Test both the k -means and k -medoid algorithms with this data set. Again, test how well the clustering corresponds to the true attributes by summarizing the results in a 3×3 matrix.